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**Trincomalee Campus, Eastern University, Sri Lanka**  
**Faculty of Applied Science**  
**Department of Computer Science**  
**Bachelor of Computer Science (2019/2020)**  
**Year I Semester II (December 2022)**  
**CO1225 Computer Architecture**

**Answer All Questions**

**Time: Two hours**

Q1)

a) Convert the following hexadecimal to

I. Binary numbers,

II. Octal numbers.

i)  $87AD_{16}$ ;

ii)  $9E7A2B_{16}$ ;

iii)  $B64FDE_{16}$ .

[12 Marks]

b) Convert the following binary numbers to decimal equivalents.

i)  $1001101_2$

ii)  $1010011.101_2$

iii)  $10101110.1001_2$

[06 Marks]

c) Convert 43.167 to binary (should be truncated to 5 digits).

[06 Marks]

d) Convert the following decimal numbers into 8-bit two's complement:

i)  $123_{10}$ ;

ii)  $-96_{10}$ .

[06 Marks]

e) Compute the following decimal arithmetic operation using 5-bit 2's complement.

i)  $5+4$

ii)  $-14+5$

iii)  $13-7$

iv)  $-8-4$

[20 Marks]

Q2)

a) Simplify the following boolean expressions using the "Boolean Algebraic" methods.

$$F = \bar{A}\bar{B}\bar{C} + ABC + A\bar{B}\bar{C} + \bar{A}BC + ABC$$

[10 Marks]

b) Truth table of a non-standard decoder as shown:

| Inputs |   |   | Outputs        |                |                |
|--------|---|---|----------------|----------------|----------------|
| A      | B | C | F <sub>0</sub> | F <sub>1</sub> | F <sub>2</sub> |
| 0      | 0 | 0 | <b>1</b>       | <b>1</b>       | <b>0</b>       |
| 0      | 0 | 1 | <b>0</b>       | <b>1</b>       | <b>1</b>       |
| 0      | 1 | 0 | <b>1</b>       | <b>0</b>       | <b>0</b>       |
| 0      | 1 | 1 | <b>0</b>       | <b>1</b>       | <b>0</b>       |
| 1      | 0 | 0 | <b>1</b>       | <b>0</b>       | <b>1</b>       |
| 1      | 0 | 1 | <b>0</b>       | <b>1</b>       | <b>1</b>       |
| 1      | 1 | 0 | <b>1</b>       | <b>0</b>       | <b>1</b>       |
| 1      | 1 | 1 | <b>1</b>       | <b>1</b>       | <b>1</b>       |

- i) Draw a block diagram for the decoder circuit showing all inputs and outputs of the Circuit. [05 Marks]
- ii) Write Boolean expression for the F<sub>0</sub>, F<sub>1</sub>, and F<sub>2</sub> outputs. [15 Marks]
- iii) Simplify the above (ii) driven equation using Boolean algebra. [10 Marks]
- iv) Draw the decoder circuit using basic logic gates. [10 Marks]

Q3)

- a) The D flip flop is mostly used in shift registers, counters and input synchronization. Draw the logic circuit and truth table for D flip flop circuit. [10 Marks]
- b) List and explain the operations performed by “Registers”. [05 Marks]
- c) Write short notes for the followings:
  - i) Memory Address Register ;
  - ii) Program Counter;
  - iii) Memory Data Register. [15 Marks]
- d) Describe the four categories of Flynn’s classification in computer architectures. [10 Marks]
- e) Briefly explain the memory hierarchy of a computer. [10 Marks]

Q4)

- a) Compare and contrast the functional operation of the “Reduced Instruction Set Architecture” (RISC) versus “Complex Instruction Set Architecture” (CISC). [09 Marks]
- b) Explain the “Instruction Cycle” in microprocessor using suitable example. [06 Marks]
- c) Explain the organization of a simple CPU with three typical buses namely, “Address Bus”, “Control Bus”, and “Data Bus” using an appropriate diagram. [08 Marks]
- d) Explain the three main Addressing Modes in computer organization. [12 Marks]
- e) Briefly describe the followings:
  - i) Virtual memory;
  - ii) Shared memory;
  - iii) Distributed memory. [15 Marks]

